

Workflow: Updating a BioCro Fork While Preserving Previous Work

This document describes a step-by-step workflow for maintaining a fork of the BioCro repository when the upstream develop branch has been updated. The procedure preserves previous work while allowing development to continue from the updated upstream code.

Situation

You forked BioCro and worked on the develop branch. Later, the upstream develop branch was updated and your changes could no longer be merged cleanly. The goal is to preserve the previous work while updating the fork to the latest upstream develop branch.

Step 1 — Navigate to the repository

```
cd E:\Sruthi\Sruthi_main\IITPKD\peaked_arrhenius_vcmax_jmax
```

Step 2 — Check repository status

```
git status
git add .
git commit -m "Save current modifications before updating develop"
```

Step 3 — Archive the current work

```
git checkout develop
git checkout -b peaked_arrhenius_archive
git push origin peaked_arrhenius_archive
```

Step 4 — Verify repository remotes

```
git remote -v
Expected remotes:
origin    -> your fork
upstream  -> https://github.com/biocro/BioCro.git
```

Step 5 — Download latest upstream updates

```
git fetch upstream
```

Step 6 — Reset local develop branch to upstream

```
git checkout develop
git reset --hard upstream/develop
```

Step 7 — Update fork on GitHub

```
git push origin develop --force
```

Step 8 — Create a new working branch

```
git checkout -b new_biocro_modifications
```

Step 9 — Continue development locally

```
git status
git add .
git commit -m "Describe modification"
```

Step 10 — Push working branch to GitHub (optional)

```
git push origin new_biocro_modifications
```

Final Repository Structure

develop	-> latest BioCro develop
new_biocro_modifications	-> current development work
peaked_arrhenius_archive	-> archived previous implementation

Access the archived work

```
git checkout peaked_arrhenius_archive
```

```
Return to current development:
git checkout new_biocro_modifications
```

Updating BioCro again in the future

```
git fetch upstream
git checkout develop
git reset --hard upstream/develop
git push origin develop --force
```